

SKIN & LASER

Laser Safety, Eyewear & Plume Control

Safety officer duties, controlled areas, wavelength-specific eyewear, fire risk and plume evacuation

DOCUMENT SKN-008	VERSION 1.0	EFFECTIVE January 2026	REVIEW CYCLE Annual
APPLIES TO All staff, including non-clinical	OWNER Medical Director	CLASSIFICATION Confidential	SUPERSEDES All prior versions

1 Purpose & Scope

This is the occupational safety document for the laser program. It covers the laser safety officer role, controlled area requirements, eyewear, fire prevention, surgical plume, device custody and the records that evidence the program exists.

A DIFFERENT REGULATOR

Laser safety sits primarily under occupational safety authority rather than your medical board — different inspectors and different penalties, and obligations you hold as an employer regardless of clinical licensure. Most practices own the eyewear and not the program.

2 Laser Safety Officer

A named individual is designated as laser safety officer in writing, with the duties below recorded and reviewed annually. The role may be held alongside another position, but it must be assigned to a person rather than assumed by the practice generally.

- ☐ Maintain the inventory of laser and light devices with wavelengths and classifications.
- ☐ Ensure wavelength-appropriate eyewear is available, in adequate quantity, and inspected.
- ☐ Establish and maintain controlled areas and signage.
- ☐ Verify operator training and authorization before independent device use.
- ☐ Maintain device logs, service records and calibration schedules.
- ☐ Control device keys and access.
- ☐ Investigate and document every incident, near miss and unexpected outcome.
- ☐ Review the program annually and after any incident or new device acquisition.

3 Eyewear

EYE INJURY IS INSTANTANEOUS AND PERMANENT

Retinal damage from a laser pulse occurs faster than the blink reflex. There is no partial compliance: if the device can fire, every person in the controlled area wears correct eyewear, including the patient, observers, sales representatives and the practice owner.

Requirement	Standard
Wavelength-specific	Eyewear must be rated for the specific wavelength in use. Protection for one platform is not protection for another. Never assume interchangeability between devices.
Optical density	The OD marked on the eyewear must meet or exceed the requirement for that wavelength and power. Verify against the device IFU.
Marked and legible	Wavelength range and OD printed on the frame and readable. Retire eyewear where the marking has worn off.
Inspected	Checked before each session for scratches, cracks, pitting or delamination. Damaged eyewear is removed from service immediately.
Quantity	Enough pairs for every person who could be in the room, plus spares.
Patient protection	External goggles for body treatment. Internal metal corneal shields where treating within the orbital rim, placed by authorized personnel with topical anesthetic per protocol.
Storage	Stored with the device it protects against, not in a shared drawer where the wrong pair can be picked up.

4 Controlled Area & Fire Safety

- ☐ Treatment room designated as a controlled area whenever the device is in use.
- ☐ Warning signage at every entrance stating the wavelength and that eyewear is required. Signage displayed only while the device is in use, so it retains meaning.
- ☐ Door closed and, where fitted, interlocked. Entry procedure defined for anyone who must enter mid-treatment, including spare eyewear immediately inside the door.
- ☐ Windows covered with appropriate barrier material.
- ☐ Reflective surfaces minimized; matte or anodized instruments used in the beam path.
- ☐ Device in standby whenever it is not actively treating — never left armed and unattended.
- ☐ No alcohol-based preparations, no flammable drapes, no supplemental oxygen in the field during ablative treatment.
- ☐ Wet gauze or drapes used around the field for ablative procedures.
- ☐ Appropriate fire extinguisher accessible and in date; staff know its location.
- ☐ Hair-bearing areas near the field wetted for ablative work; singeing hair is an ignition source.

IGNITION RISK IS REAL

Operating room fires involving lasers are documented and devastating. The three ingredients are an ignition source, fuel, and an oxidizer. You control the fuel — alcohol preps, dry drapes, hair — and the oxidizer. Never combine supplemental oxygen with an ablative laser on the face.

5 Surgical Plume

Plume generated by ablative laser, RF and electrosurgical devices contains particulate matter, chemical byproducts and potentially viable cellular and viral material. It is an occupational exposure for staff, who encounter it repeatedly across a working week.

- ☐ Local smoke evacuation used for all plume-generating procedures, with the capture nozzle held close to the point of generation — effectiveness falls sharply with distance.
- ☐ Evacuator filters replaced on the manufacturer's schedule and the replacement logged.
- ☐ Filters handled as contaminated waste when changed, with gloves and eye protection.
- ☐ Respiratory protection appropriate to the procedure worn by everyone in the room; a standard procedure mask does not filter plume particulate.
- ☐ Room ventilation adequate; general HVAC is not a substitute for local evacuation.
- ☐ Eye protection worn — plume is an ocular irritant in addition to the laser hazard.
- ☐ Staff informed of plume risks as part of laser safety training, and the training documented.

6 Device Custody, Logs & Records

- Device keys secured and controlled by the laser safety officer, released only to authorized operators.
- Device use log recording date, operator, patient identifier, treatment area and parameters.
- Manufacturer service and calibration schedule followed, with records retained.
- Any fault, error code or unexpected output logged and the device removed from service until resolved.
- Operator authorization cross-referenced to the competency matrix in OPS-003.
- Annual laser safety training for all staff, including non-clinical staff who may enter the controlled area, with attendance recorded.
- Incident and near-miss reports retained with investigation outcomes.
- Annual program review documented, and a review triggered by any new device.

INSPECTION-READY TEST

An inspector should be able to ask for the laser safety officer by name, the device inventory, the eyewear inspection records, the training log and the device use logs — and receive all five within a few minutes. If that is not true today, that is the first corrective action.

7 Quick Reference

QUICK REFERENCE CARD

Before the device is armed

1. Door closed. Sign up. Windows covered.

2. Correct-wavelength eyewear on everyone — including observers and reps.

3. Eyewear inspected for scratches and legible OD marking.

4. Patient shields — external goggles, or corneal shields within the orbital rim.

5. Ablative: no alcohol prep, no oxygen, wet drapes, extinguisher located.

6. Plume evacuation on, nozzle close to the source, correct respiratory protection.

7. Standby whenever not treating. Never armed and unattended.

8. Log the session. Key back to the safety officer.

Print, laminate and post at the point of care

8 Medical Director Authorization

This protocol is adopted as a standing order for the practice named below. By signing, the medical director confirms the protocol has been reviewed against current clinical guidance, applicable state scope-of-practice rules, and the training level of the staff authorized to perform it.

PRACTICE NAME

LOCATION / SITE

MEDICAL DIRECTOR — PRINTED NAME & CREDENTIALS

LICENSE NUMBER & STATE

SIGNATURE

DATE ADOPTED / NEXT REVIEW DATE

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